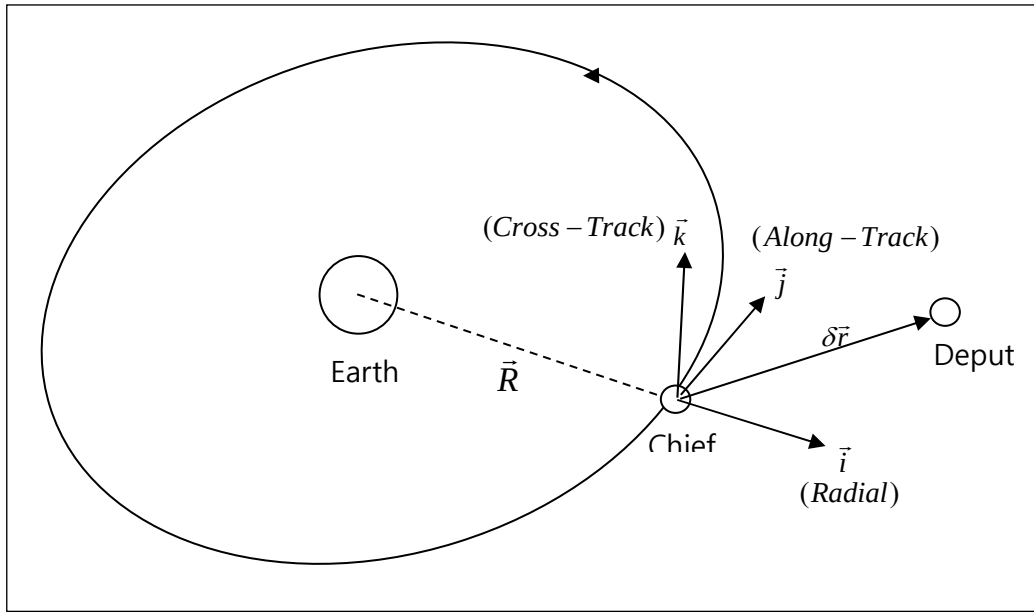


AST 6007-01 Spacecraft Dynamics and Control

Homework #1

Due Sep 22 2021, 10:00AM

1. (Equations of Motion for Relative Motion with respect to a Circular Reference Orbit)



Consider a local-vertical-local-horizontal xyz-frame attached to a circular reference orbit whose constant orbital frequency is given by $\vec{\omega} = \omega \vec{k}$. Let the position of deputy spacecraft be given by

$$\vec{r} = \vec{R} + \delta\vec{r} = (R + x)\vec{i} + y\vec{j} + z\vec{k}$$

- a. Show that the velocity and acceleration vector of the deputy spacecraft are

$$\begin{aligned}\dot{\vec{r}} &= [\dot{x} - \omega y]\vec{i} + [\dot{y} + \omega(R + x)]\vec{j} + \dot{z}\vec{k} \\ \ddot{\vec{r}} &= [\ddot{x} - 2\omega\dot{y} - \omega^2(R + x)]\vec{i} + [\ddot{y} + 2\omega\dot{x} - \omega^2 y]\vec{j} + \ddot{z}\vec{k}\end{aligned}$$

- b. Using the Newton's Law

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{u}$$

derive the equations of motion in terms of $(\vec{i}, \vec{j}, \vec{k})$ -component. Here μ is the gravitational parameter of the Earth, $\vec{u} = [u_x \ u_y \ u_z]^T$ is the thrust acceleration vector, and $r = \sqrt{(R + x)^2 + y^2 + z^2}$ is the distance of the deputy spacecraft from the Earth center.

- c. Non-dimensionalize the equations of motion in (b) by using the canonical units, i.e., by using the reference length R and the reference time $1/\omega$.
- d. Linearize the equations of motion in canonical units with respect to the circular reference trajectory to obtain Clohessy-Wiltshire equations of motion.

Consider the 2-body model for a spacecraft in low Earth orbit. The spacecraft orbit is determined by the gravity of the Earth and by propulsive forces on the spacecraft that are expressed in the LVLH coordinate frame.

2. What are the nonlinear differential equations, expressed in spherical coordinates, which describe the dynamics of the spacecraft under the influence of the gravity of the Earth and propulsive forces? Use the specific value of μ corresponding to Earth orbits.
3. Suppose that the spacecraft is in a $600km$ altitude circular orbit in a fixed orbital plane corresponding to $\lambda = 0$, with no propulsive forces acting on it.
 - a. What is the constant orbital velocity of the spacecraft?
 - b. What is the orbital period of the spacecraft?
4. Consider perturbations of the spacecraft from the $600km$ altitude circular orbit. What are the linear differential equations (the Clohessy-Wiltshire equations) that describe the perturbation dynamics of the spacecraft under the influence of the gravity of Earth and propulsive forces?
5. Suppose that the spacecraft is initially on a $600km$ altitude circular orbit but its orbital velocity is instantaneously perturbed: the direction of the orbital velocity is not changed but its magnitude is increased by 1%. Based on the linear perturbation dynamics in Problems 4:
 - a. What are the orbital position vector and the orbital velocity vector during one orbital revolution assuming there are no propulsive forces?
 - b. Describe the orbit that results from this initial perturbation by giving its eccentricity, its radius of perigee, its radius of apogee, and its orbital period.
6. Using the linear differential equations in Problem 3, assess the following state controllability properties of the spacecraft and interpret your conclusions:
 - a. Are the spacecraft in-plane dynamics completely controllable using only local vertical propulsive forces F_r ?
 - b. Are the spacecraft in-plane dynamics completely controllable using only local horizontal propulsive forces F_v ?
 - c. Is the spacecraft completely controllable using local vertical propulsive forces F_r and local horizontal propulsive forces F_v ?
7. Design a rendezvous maneuver between two spacecraft. Assume the first spacecraft is initially in a $580km$ altitude circular orbit and the second spacecraft is in a $600km$ altitude circular orbit; assuming the two orbits

are co-planar, and at the initial time their spacecraft positions are aligned along a common radial line. The first spacecraft is to execute a rendezvous maneuver with the second spacecraft requiring exactly 30 minutes to complete.

- a. Describe a rendezvous maneuver for the first spacecraft that uses two propulsive force impulses, assuming the first impulse occurs when the maneuver is initiated and the second impulse occurs when the maneuver is terminated. For each impulse, describe the direction and magnitude of the instantaneous change in the first spacecraft velocity vector that achieve the desired result. Verify through analysis or simulation that the rendezvous objective is achieved using your propulsive force scheme.
- b. Describe a rendezvous maneuver for the first spacecraft that uses a non-impulsive propulsive force. Describe the direction and magnitude of local vertical and local horizontal propulsive forces that achieve the desired result. Verify through analysis or simulation that the rendezvous objective is achieved using your propulsive force scheme.